

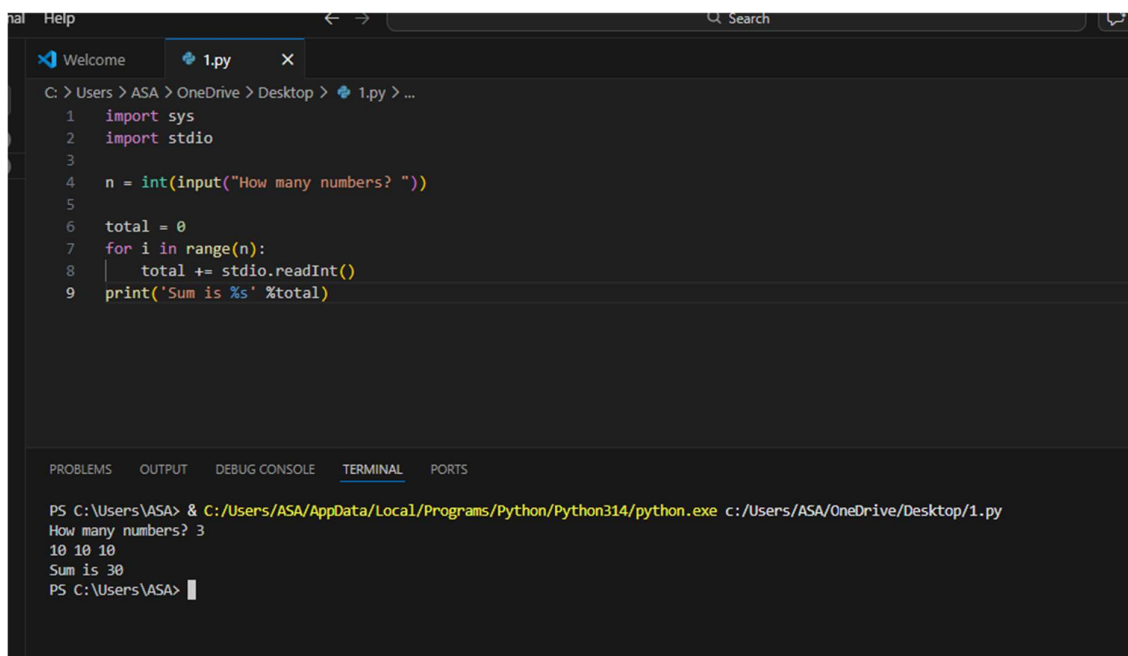
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تکلیف جلسه ششم . درس برنامه نویسی پیشرفته

تمرین شماره 1 .

دلیل استفاده از %F به جای %S در کد زیر :



```
1 import sys
2 import stdio
3
4 n = int(input("How many numbers? "))
5
6 total = 0
7 for i in range(n):
8     total += stdio.readInt()
9 print('Sum is %s' %total)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/1.py
How many numbers? 3
10 10 10
Sum is 30
PS C:\Users\ASA>
```

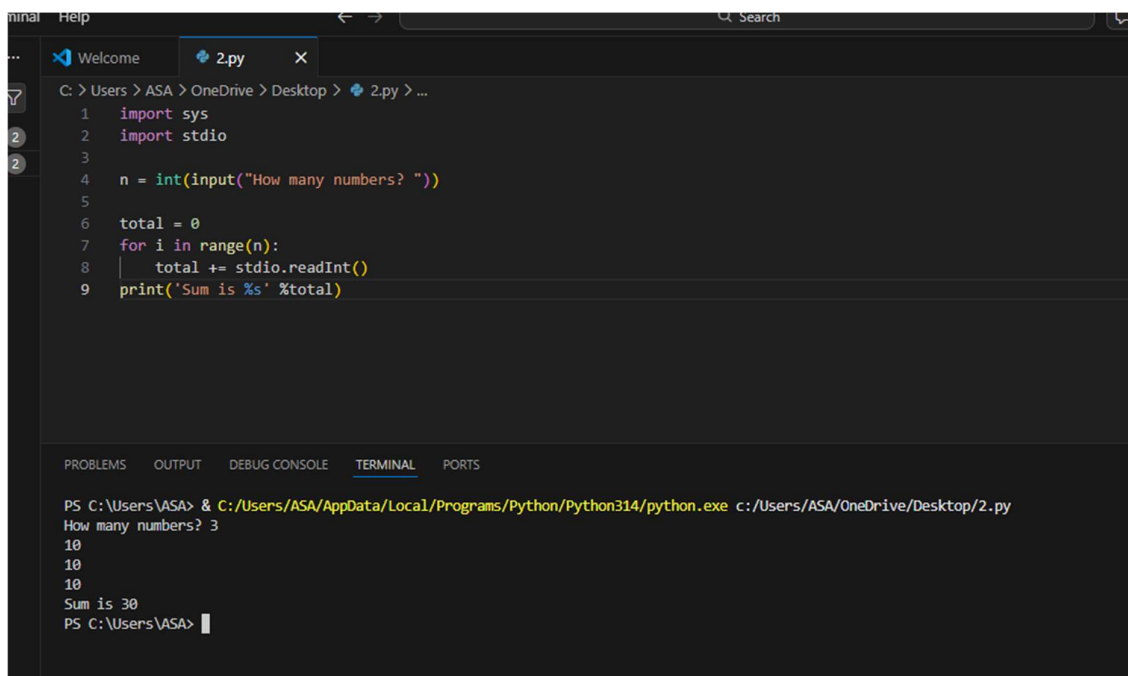
%f مخصوص اعداد اعشاری (Float) است

%s مخصوص رشته‌ها (String) است: وقتی از '%s' استفاده می‌کنیم، پایتون به صورت خودکار هر چیزی را که به او بدهیم ابتدا به متن (String) تبدیل می‌کند و بعد نمایش می‌دهد.

تمرین شماره 2 .

در تمرین قبل اعداد به شکل ردیفی نوشته شدن اما در این تمرین میخواهیم اعداد به شکل ستونی نوشته شوند . چه کار باید کرد ؟

ما در مثال قبل در بخش عدد دهی یک اعداد را پشت سر هم وارد می کردیم اما در این تمرین نیاز هست که هر عدد را وارد کنیم سپس enter بزنیم و عدد بعدی را می نویسم



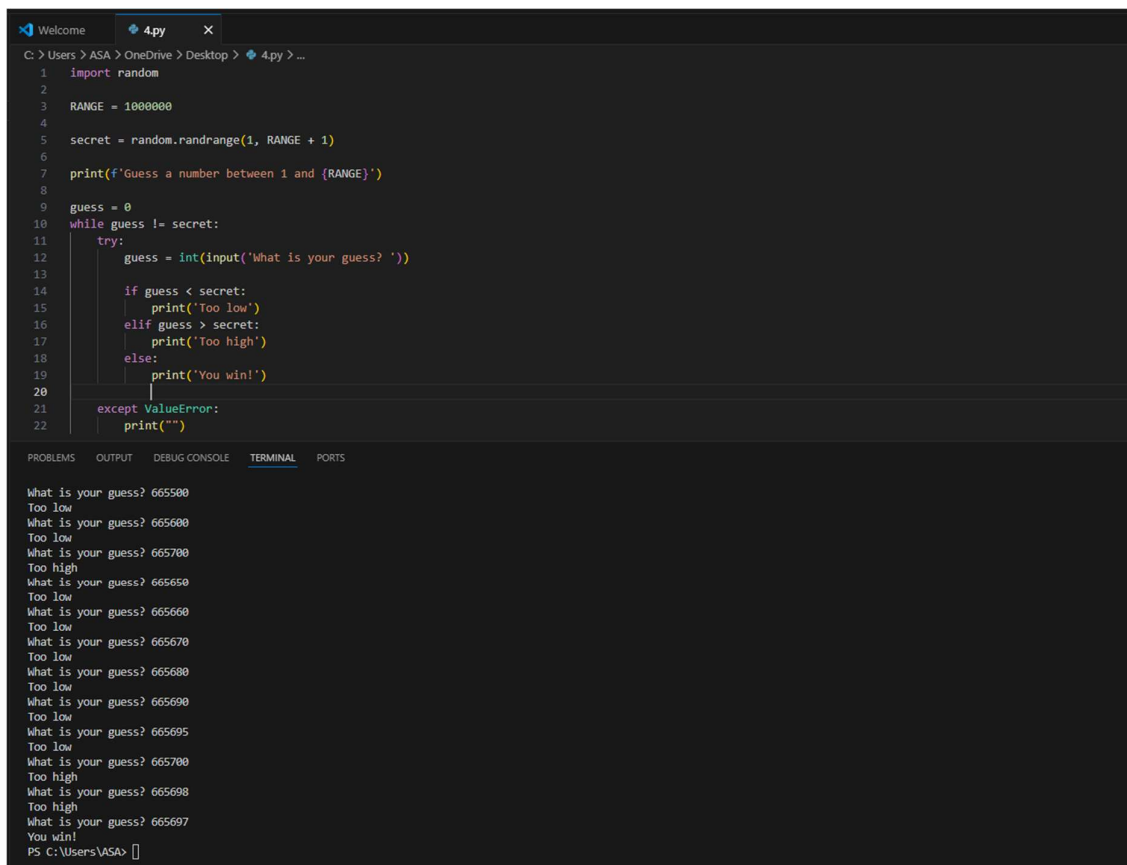
```
1 import sys
2 import stdio
3
4 n = int(input("How many numbers? "))
5
6 total = 0
7 for i in range(n):
8     total += stdio.readInt()
9 print('Sum is %s' %total)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/2.py
How many numbers? 3
10
10
10
Sum is 30
PS C:\Users\ASA>
```

تمرین شماره 3 .

ورودی خروجی تعاملی :



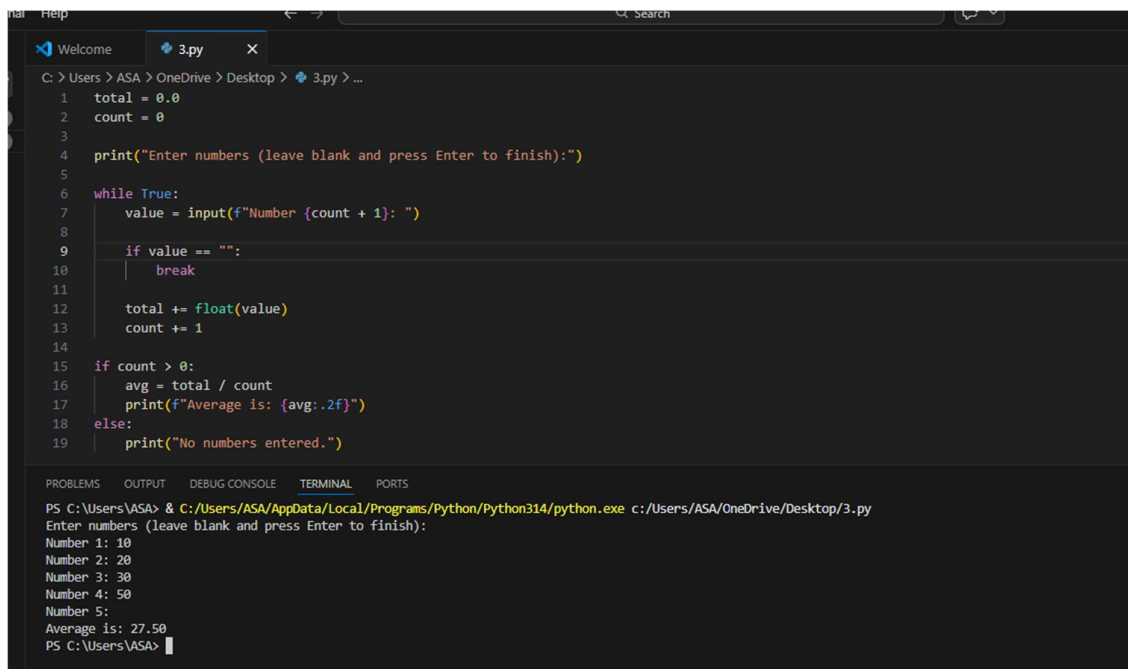
```
1 import random
2
3 RANGE = 1000000
4
5 secret = random.randrange(1, RANGE + 1)
6
7 print(f'Guess a number between 1 and {RANGE}')
8
9 guess = 0
10 while guess != secret:
11     try:
12         guess = int(input('What is your guess? '))
13
14         if guess < secret:
15             print('Too low')
16         elif guess > secret:
17             print('Too high')
18         else:
19             print('You win!')
20     except ValueError:
21         print("")
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
What is your guess? 665500
Too low
What is your guess? 665600
Too low
What is your guess? 665700
Too high
What is your guess? 665650
Too low
What is your guess? 665660
Too low
What is your guess? 665670
Too low
What is your guess? 665680
Too low
What is your guess? 665690
Too low
What is your guess? 665695
Too low
What is your guess? 665700
Too high
What is your guess? 665698
Too high
What is your guess? 665697
You win!
PS C:\Users\ASAD>
```

تمرین شماره 4 .

میانگین گیری اعداد :



The screenshot shows a Python IDE with a file named '3.py' open. The code in the editor is as follows:

```
1 total = 0.0
2 count = 0
3
4 print("Enter numbers (leave blank and press Enter to finish):")
5
6 while True:
7     value = input(f"Number {count + 1}: ")
8
9     if value == "":
10        break
11
12    total += float(value)
13    count += 1
14
15 if count > 0:
16     avg = total / count
17     print(f"Average is: {avg:.2f}")
18 else:
19     print("No numbers entered.")
```

The bottom panel of the IDE shows the terminal output:

```
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/3.py
Enter numbers (leave blank and press Enter to finish):
Number 1: 10
Number 2: 20
Number 3: 30
Number 4: 50
Number 5:
Average is: 27.50
PS C:\Users\ASA>
```

تمرین شماره 5 .

تغییر مسیر ورودی :

```

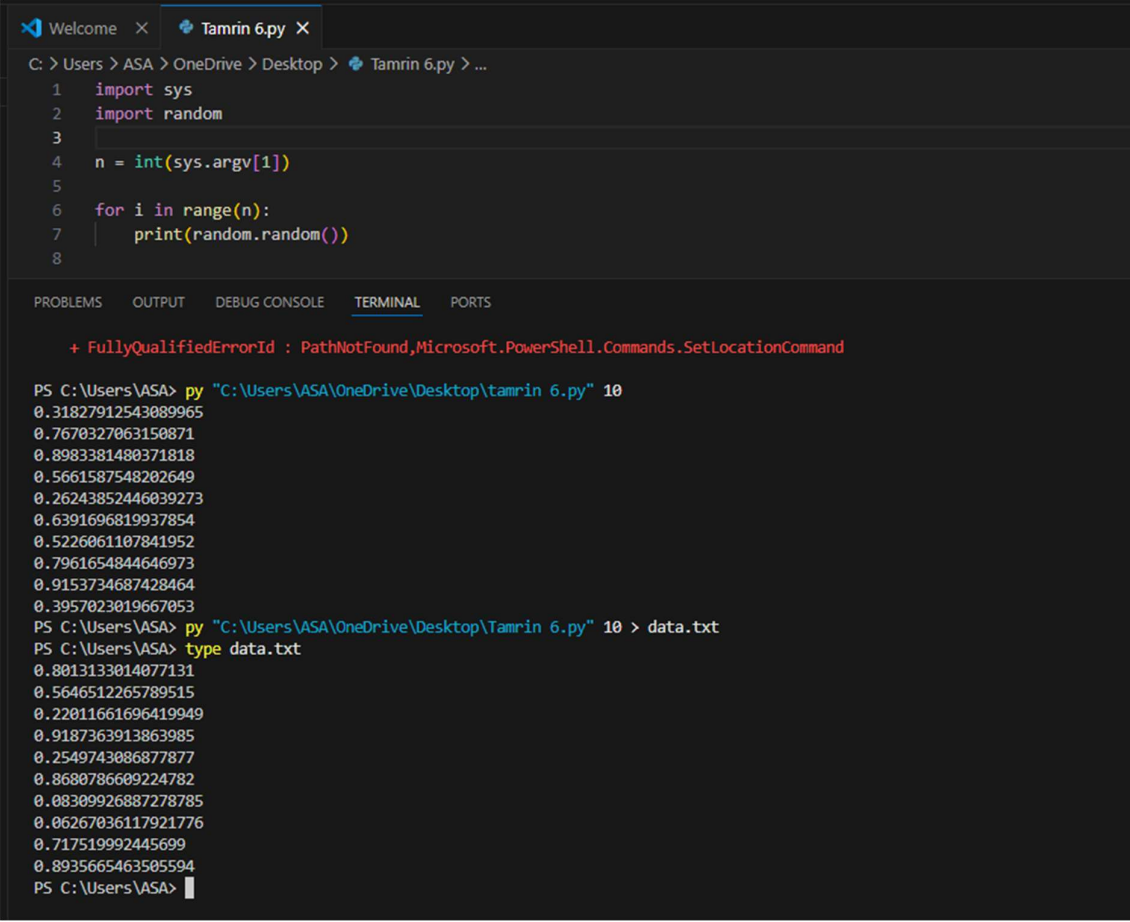
Welcome  vorodi.py X
C: > Users > ASA > OneDrive > Desktop > vorodi.py > ...
1  import sys
2
3  total = 0
4  count = 0
5
6  for line in sys.stdin:
7      try:
8          total += float(line.strip())
9          count += 1
10     except ValueError:
11         continue
12
13 if count > 0:
14     print(f"Average: {total/count}")
15
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
powerShell + v []
>> try:
>>     total += float(line.strip())
>>     count += 1
>> except ValueError:
>>     continue # بدون ددع ریگذا
>>
>> if count > 0:
>>     print(f"Average: {total/count}")
>>
At line:6 char:4
+ for line in sys.stdin:
+ ~
Missing opening '(' after keyword 'for'.
At line:8 char:35
+         total += float(line.strip())
+ ~
An expression was expected after '('.
At line:13 char:3
+ if count > 0:
+ ~
Missing '(' after 'if' in if statement.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : MissingOpenParenthesisAfterKeyword

PS C:\Users\ASA> py "C:\Users\ASA\OneDrive\Desktop\vorodi.py" < data.txt
At line:1 char:46
+ py "C:\Users\ASA\OneDrive\Desktop\vorodi.py" < data.txt
+ ~
The '<' operator is reserved for future use.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : RedirectionNotSupported

PS C:\Users\ASA> Get-Content data.txt | py
PS C:\Users\ASA> C:\Users\ASA\OneDrive\Desktop\vorodi.py
PS C:\Users\ASA>
```

تمرین 6 .

تغییر مسیر خروجی :



The image shows a Visual Studio Code editor window with a Python file named 'Tamrin 6.py' open. The code in the file is a simple script that takes a command-line argument 'n' and prints 'n' random floating-point numbers between 0 and 1. The terminal at the bottom shows the command to run the script with 'n=10' and the resulting output of 10 random numbers. A red error message is also visible in the terminal output.

```
1 import sys
2 import random
3
4 n = int(sys.argv[1])
5
6 for i in range(n):
7     print(random.random())
8
```

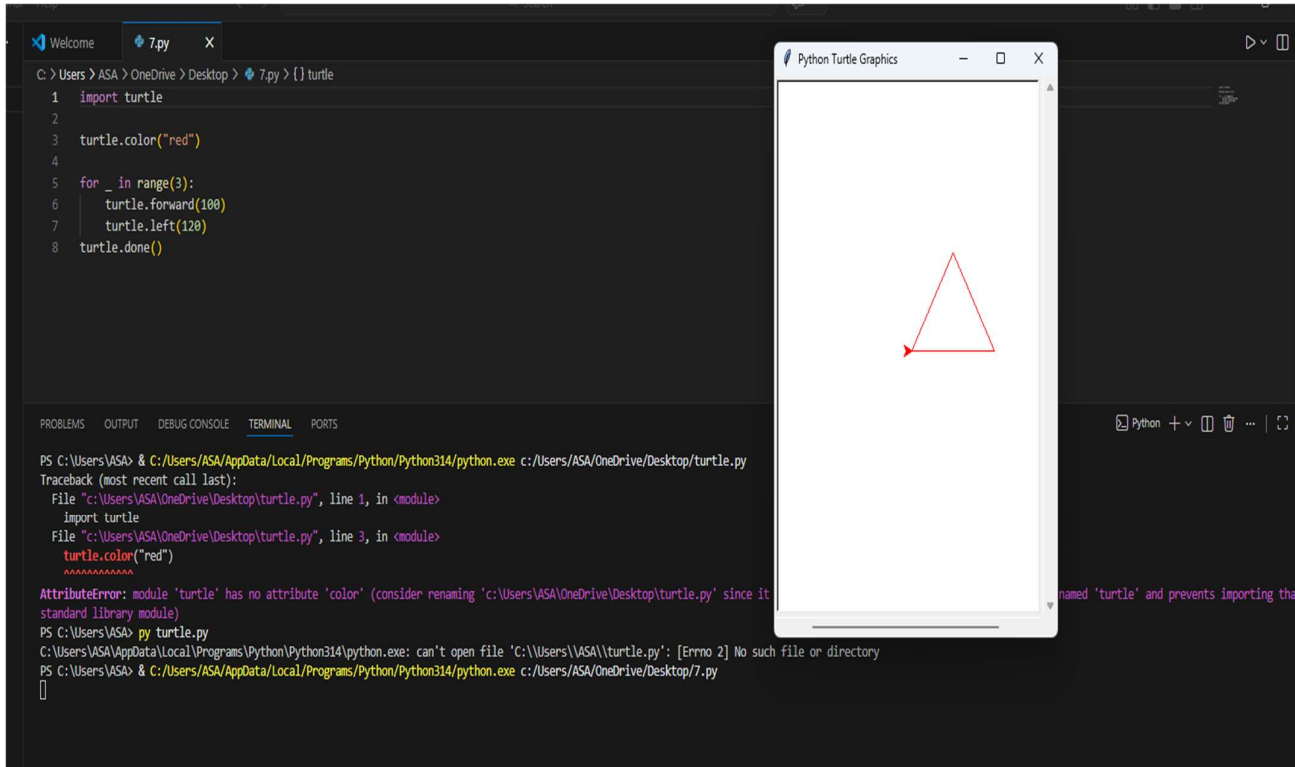
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\Users\ASA> py "C:\Users\ASA\OneDrive\Desktop\tamrin 6.py" 10
0.318279125430889965
0.7670327063150871
0.8983381480371818
0.5661587548202649
0.26243852446039273
0.6391696819937854
0.5226061107841952
0.7961654844646973
0.9153734687428464
0.3957023019667053
PS C:\Users\ASA> py "C:\Users\ASA\OneDrive\Desktop\tamrin 6.py" 10 > data.txt
PS C:\Users\ASA> type data.txt
0.8013133014077131
0.5646512265789515
0.22011661696419949
0.9187363913863985
0.2549743086877877
0.8680786609224782
0.08309926887278785
0.06267036117921776
0.717519992445699
0.8935665463505594
PS C:\Users\ASA>

تمرین 7 .

مثال لاکپشت :



The screenshot shows a Python IDE with a file named `7.py` open. The code in the editor is as follows:

```
1 import turtle
2
3 turtle.color("red")
4
5 for _ in range(3):
6     turtle.forward(100)
7     turtle.left(120)
8 turtle.done()
```

Below the editor, the terminal window displays the following output:

```
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/turtle.py
Traceback (most recent call last):
  File "c:\Users\ASA\OneDrive\Desktop\turtle.py", line 1, in <module>
    import turtle
  File "c:\Users\ASA\OneDrive\Desktop\turtle.py", line 3, in <module>
    turtle.color("red")
AttributeError: module 'turtle' has no attribute 'color' (consider renaming 'c:\Users\ASA\OneDrive\Desktop\turtle.py' since it
standard library module)
PS C:\Users\ASA> py turtle.py
C:\Users\ASA\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'C:\Users\ASA\turtle.py': [Errno 2] No such file or directory
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/7.py
```

On the right side of the IDE, a window titled "Python Turtle Graphics" is open, displaying a red equilateral triangle on a white background. The triangle is drawn with a red line and has a small red arrow at its bottom-left vertex, indicating the starting point and direction of the turtle.

تمرین 8 :

بازی اشوب : مثلث سیرپینسکی

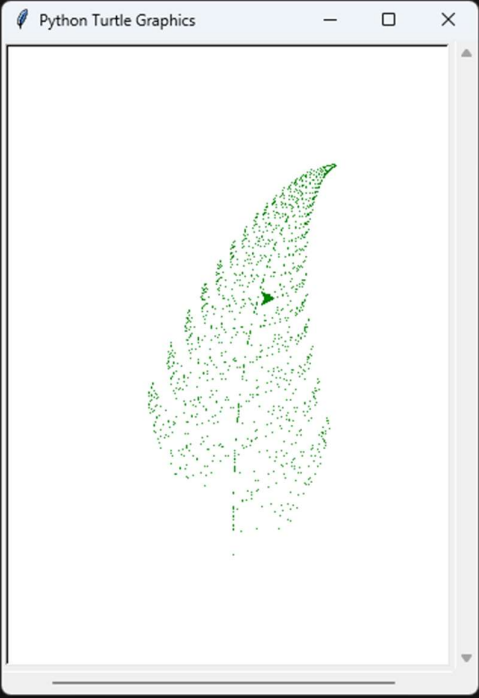
```
mininal Help 1/4 Search
... Welcome 8.py x
C: > Users > ASA > OneDrive > Desktop > 8.py > ...
1 import turtle
2 import random
3
4 screen = turtle.Screen()
5 t = turtle.Turtle()
6 t.speed(0)
7 t.hideturtle()
8 t.penup()
9
10
11 vertices = [
12     (0, 200, "blue"),
13     (-200, -200, "red"),
14     (200, -200, "green")
15 ]
16
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/8.py
Python Turtle Graphics

```


تمرین 9 :

سرخس بارنزلی :

```

Welcome 9.py x
C: > Users > ASA > OneDrive > Desktop > 9.py > _
1 import turtle
2 import random
3
4 t = turtle.Turtle()
5 t.speed(0)
6 t.color("green")
7 t.penup()
8
9 x = 0.0
10 y = 0.0
11
12 for _ in range(3000):
13     t.goto(x * 30, y * 30 - 150)
14     t.dot(2)
15
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/9.py
Python Turtle Graphics

```

تمرین 10 :

صخره مرجانی :

```
1 import turtle
2 import random
3
4 t = turtle.Turtle()
5 t.speed(0)
6 t.penup()
7 t.color("brown")
8
9 x = 0.0
10 y = 0.0
11
12 for i in range(5000):
13     px = x.__mul__(100)
14     py = y.__mul__(100).__sub__(150)
15
16     t.goto(px, py)
17     t.dot(2)
18
19     r = random.random()
20
21     if r < 0.1:
22         nx = x.__mul__(0.05)
23         ny = y.__mul__(0.6)
24     elif r < 0.2:
25         nx = x.__mul__(0.05)
26         ny = y.__mul__(-0.5).__add__(1.0)
27     elif r < 0.6:
28         nx = x.__mul__(0.46).__sub__(y.__mul__(0.32))
29         ny = x.__mul__(0.39).__add__(y.__mul__(0.38)).__add__(0.6)
30     else:
31         nx = x.__mul__(0.47).__sub__(y.__mul__(0.15))
32         ny = x.__mul__(0.17).__add__(y.__mul__(0.42)).__add__(1.1)
33
34     x = nx
35     y = ny
36
37 turtle.done()
```

